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RESEARCH INTERESTS

Non-genetic neuromodulation with photothermal nanoparticles · Multifunctional fiber neural probes by thermal drawing · Nanoparticle surface chemistry for membrane anchoring and cell-type targeting · Memory engram allocation and control · Experience-independent concept transfer

EDUCATION

Sep 2020 — Aug 2026 **Ph.D. in Brain and Cognitive Engineering**
Korea Advanced Institute of Science and Technology (KAIST)
Integrated M.S.–Ph.D. program, advised by Prof. Seongjun Park

Mar 2015 — Aug 2020 **B.S. in Electrical Engineering and Bio and Brain Engineering**
Korea Advanced Institute of Science and Technology (KAIST)
Double major

RESEARCH EXPERIENCE

Sep 2026 — present **Postdoctoral Researcher**
Medical Research Center, Seoul National University
Non-genetic neuromodulation and memory engram control

Sep 2020 — Aug 2026 **Doctoral Researcher**
Bio and Neural Interfaces Laboratory, KAIST
Advised by Prof. Seongjun Park

- Built multifunctional fiber neural probes by thermal drawing, integrating a microfluidic channel, an optical waveguide and recording electrodes in one filament, and characterized their optical, fluidic, electrical and mechanical properties
- Synthesized gold nanorods and exchanged their surface ligands for cholesterol-terminated PEG to make them biocompatible and membrane-anchoring
- Carried out stereotaxic implantation and in vivo single- and multi-unit recording in mouse hippocampus and entorhinal cortex
- Established the photothermal stimulation parameters that separate inhibition from excitation, and assessed tissue response by NeuN and Iba1 immunohistochemistry

PUBLICATIONS

2026 **Nongenetic in Vivo Bimodal Neuromodulation via Photothermal Gold Nanorods and a Multifunctional Fiber Neural Probe**
C. Yoon, Y. Lee, G. Yim, S.-W. Lee, W. Lee, Y.-G. Park, S. Park
ACS Nano 20(28), 20001–20015
[doi:10.1021/acs.nano.6c02201](https://doi.org/10.1021/acs.nano.6c02201)

2026 **Thermally Drawn Multifunctional All-Hydrogel Fibers for Anti-Fibrotic and Multimodal Neural Interfaces**
C. Sung, K. S. Nam, Y. Kim, H. Kang, K. Kim, **C. Yoon**, S. Lee, A. Chung, J. Kang, Y.-G. Park, A. J. Park, H. Butt, H. Yuk, S. Park
Advanced Materials 38(9), e11634
[doi:10.1002/adma.202511634](https://doi.org/10.1002/adma.202511634)

2022

A Personalized Electronic Tattoo for Healthcare Realized by On-the-Spot Assembly of an Intrinsically Conductive and Durable Liquid-Metal Composite

G.-H. Lee, H. Woo, **C. Yoon**, C. Yang, J.-Y. Bae, W. Kim, D. H. Lee, H. Kang, S. Han, S.-K. Kang, S. Park, H.-R. Kim, J.-W. Jeong, S. Park

Advanced Materials 34(32), 2204159

doi:10.1002/adma.202204159

2021

Recent Advances in Recording and Modulation Technologies for Next-Generation Neural Interfaces

J.-W. Hong, **C. Yoon**, K. Jo, J. H. Won, S. Park

iScience 24(12), 103550 · Review

doi:10.1016/j.isci.2021.103550

CONFERENCE PRESENTATIONS

2026

Non-Genetic Engram Allocation-and-Manipulation in Mouse Brain via Gold-Nanorod Mediated Photothermal Stimulation

Korean Society for Brain and Neural Sciences (KSBNS)

Poster

2025

Non-Genetic In Vivo Bimodal Neuromodulation Enabled by Fiber-Based Neural Probe and Gold Nanorod

Society for Neuroscience (SfN)

Poster

2025

Non-Genetic In Vivo Bimodal Neuromodulation Enabled by Fiber-Based Neural Probe and Gold Nanorod

IEEE International Conference on Neural Engineering (NER)

Poster

2023

Non-Genetic Bimodal In Vivo Neuromodulation Using Gold Nanorod-Mediated Photothermal Stimulation

Korean Society for Brain and Neural Sciences (KSBNS)

Poster

2022

Non-Genetic, Bidirectional In Vivo Photothermal Neuromodulation Enabled by Gold Nanorod and Fiber-Based Multifunctional Neural Interface

Korean BioChip Society (KBSC)

Poster

SKILLS AND TECHNIQUES

Fabrication	Thermal drawing of multifunctional fibers, Preform design, Neural probe assembly
Nanomaterials	Seed-mediated gold nanorod synthesis, Ligand exchange (CTAB to cholesterol-PEG), TEM, UV-Vis spectroscopy, zeta potential
In vivo	Stereotaxic implantation surgery, Single- and multi-unit electrophysiology, Photothermal neuromodulation, Immunohistochemistry (NeuN, Iba1)
Characterization	Optical attenuation, Impedance spectroscopy, Bending stiffness, Infrared thermography
Computation	MATLAB, Python, SolidWorks, Bioheat modeling, Spike sorting and PETH analysis